

alfevering 8

2025-04-06

dataindlæsning

```
oktan<-read.csv("/Users/asgeraircool/rdata/oktan.csv")
head(oktan)
```

```
##           T1           T2           T3           T4           T5 oktan
## 1 0.292567 0.466098 0.044695 0.017254 0.311237 85.30
## 2 0.316834 0.516133 0.056740 0.020912 0.342958 85.25
## 3 0.276645 0.504049 0.055821 0.022470 0.352530 88.45
## 4 0.319126 0.497873 0.052995 0.022399 0.325323 83.40
## 5 0.251579 0.432696 0.045830 0.018535 0.320695 87.90
## 6 0.295713 0.478875 0.048868 0.019856 0.320822 85.50
```

##a

$M_0 : oktan_i \sim N(\alpha + \beta_{T1} * T1_i + \beta_{T2} * T2_i + \beta_{T3} * T3_i + \beta_{T4} * T4_i + \beta_{T5} * T5_i, \sigma_0^2) \text{ for } i \in \{1, \dots, n_i\}$

##b

vi kører backwards selection proceduren

```
m0<-lm(oktan~T1+T2+T3+T4+T5,data=oktan)
summary(m0)
```

```
##
## Call:
## lm(formula = oktan ~ T1 + T2 + T3 + T4 + T5, data = oktan)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.70559 -0.16640  0.00369  0.14092  0.69024
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   91.225      2.364  38.584 < 2e-16 ***
## T1            -89.771       6.123 -14.661 < 2e-16 ***
## T2              30.505      10.096   3.021  0.00384 **
## T3            -37.670      40.097  -0.939  0.35168
## T4              28.279      29.113   0.971  0.33570
## T5              22.732      16.322   1.393  0.16941
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2732 on 54 degrees of freedom
## Multiple R-squared:  0.9708, Adjusted R-squared:  0.9681
## F-statistic: 359.2 on 5 and 54 DF,  p-value: < 2.2e-16
```

$\beta_{t3} = 0$ har størst p værdi da går den ud som den første

```
m1<-lm(oktan~T1+T2+T4+T5,data=oktan)
summary(m1)
```

```
##
## Call:
## lm(formula = oktan ~ T1 + T2 + T4 + T5, data = oktan)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.73920 -0.15023  0.00228  0.15407  0.68630
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   93.053      1.342   69.316 < 2e-16 ***
## T1            -92.543      5.360  -17.267 < 2e-16 ***
## T2             32.227      9.918   3.249  0.00197 **
## T4              3.975     13.339   0.298  0.76683
## T5             12.904     12.516   1.031  0.30703
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2729 on 55 degrees of freedom
## Multiple R-squared:  0.9703, Adjusted R-squared:  0.9682
## F-statistic: 449.8 on 4 and 55 DF,  p-value: < 2.2e-16
```

nu T4

```
m2<-lm(oktan~T1+T2+T5,data=oktan)
summary(m2)
```

```
##
## Call:
## lm(formula = oktan ~ T1 + T2 + T5, data = oktan)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.73731 -0.14391 -0.00221  0.15662  0.67330
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   92.7552      0.8892 104.314 < 2e-16 ***
## T1            -91.5582      4.1856 -21.875 < 2e-16 ***
## T2             30.5806      8.1686   3.744  0.00043 ***
## T5             15.5788      8.6521   1.801  0.07715 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2707 on 56 degrees of freedom
## Multiple R-squared:  0.9703, Adjusted R-squared:  0.9687
## F-statistic: 609.6 on 3 and 56 DF,  p-value: < 2.2e-16
```

nu T5

```
m3<-lm(oktan~T1+T2,data=oktan)
summary(m3)
```

```
##
## Call:
## lm(formula = oktan ~ T1 + T2, data = oktan)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.78638 -0.16527  0.01215  0.17415  0.58995
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  92.8562     0.9047  102.64  <2e-16 ***
## T1          -97.6559     2.5076  -38.94  <2e-16 ***
## T2           44.5063     2.6802   16.61  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.276 on 57 degrees of freedom
## Multiple R-squared:  0.9686, Adjusted R-squared:  0.9675
## F-statistic: 878.3 on 2 and 57 DF,  p-value: < 2.2e-16
```

hvorefter proceduren stopper og vi ender i model 3

$$M_3 : oktan_i \sim N(\alpha + \beta_{T1} * T1_i + \beta_{T2} * T2_i, \sigma_0^2) \text{ for } i \in \{1, \dots, n_i\}$$

##c

vi tjekker om reductionen er rimelig med anova

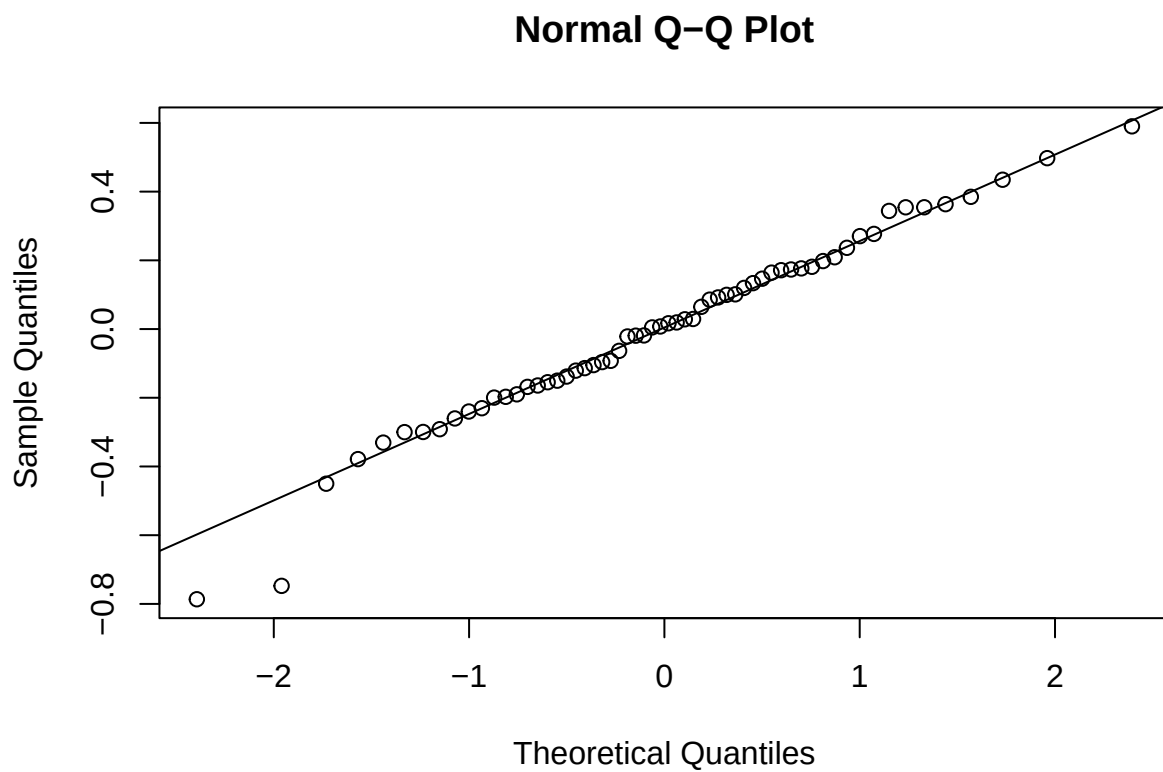
```
anova(m3,m0)
```

```
## Analysis of Variance Table
##
## Model 1: oktan ~ T1 + T2
## Model 2: oktan ~ T1 + T2 + T3 + T4 + T5
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      57 4.3414
## 2      54 4.0313  3    0.3101 1.3846 0.2574
```

som har en P værdi over 0.05 da er reductionen rimelig

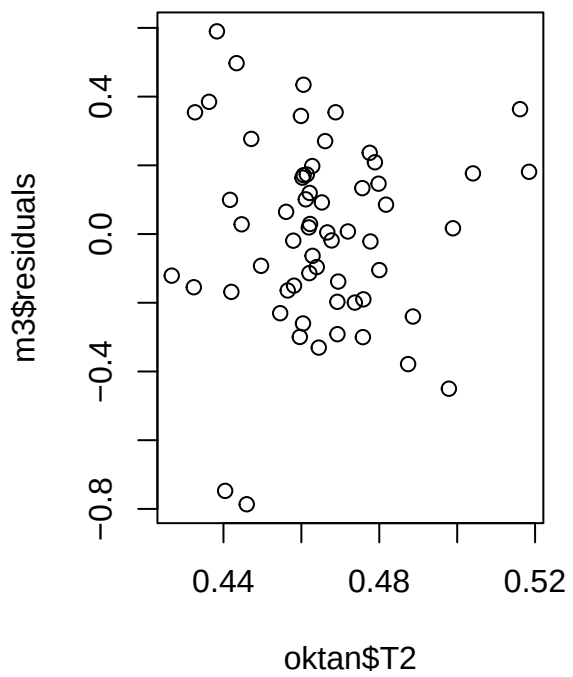
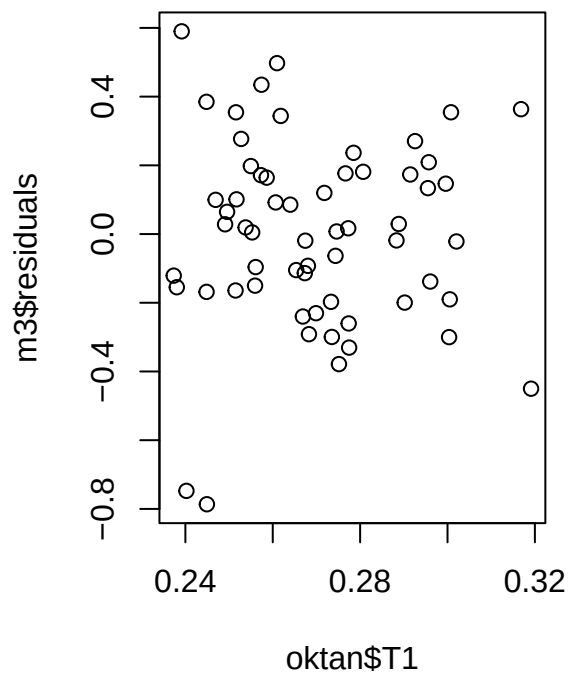
##d

```
qqnorm(m3$residuals)
qqline(m3$residuals)
```

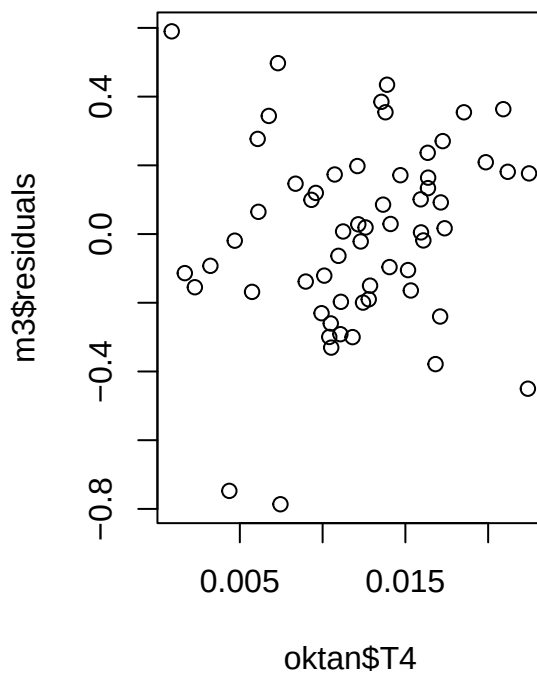
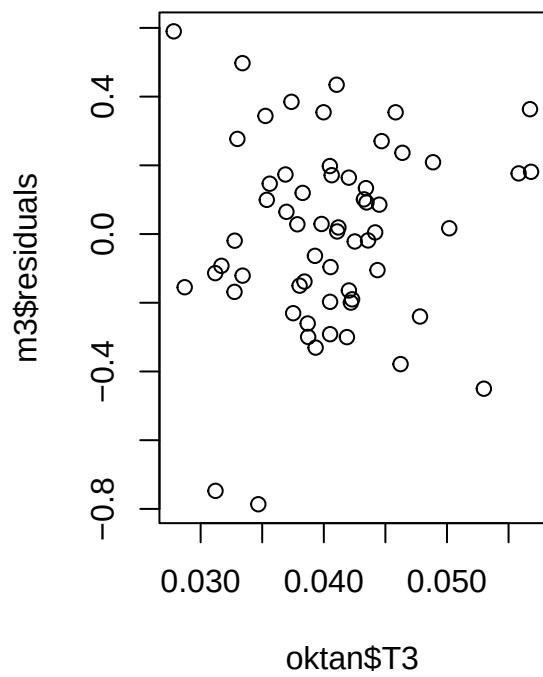


vi har en super fin snoen ad qq line da kan residualerne antages normalfordelte vi plotter også residualerne mod de forskellige variable

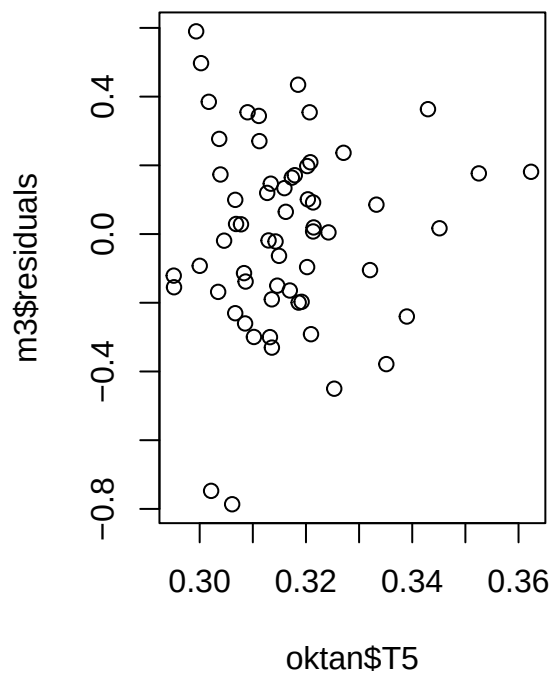
```
par(mfrow=c(1,2))  
plot(m3$residuals~oktan$T1)  
plot(m3$residuals~oktan$T2)
```



```
plot(m3$residuals~oktan$T3)
plot(m3$residuals~oktan$T4)
```



```
plot(m3$residuals~oktan$T5)
```



vi ser også på de 5 residual plots at det ikke ligner at der er nogen systematisk afvigelse i residualerne
 ##e vi aflæser blot fra et konfint kald

```
confint(m3)
```

```
##                2.5 %    97.5 %
## (Intercept)   91.04452  94.66783
## T1            -102.67741 -92.63447
## T2             39.13918  49.87338
```

for de forskellige α [91.04452 , 94.66783] β_{T1} [-102.67741 , -92.63447] β_{T2} [39.13918 , 49.87338]